

VENKATESH P

AI/ML Engineer · Generative AI · MLOps · LLM Systems

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Chennai, India · Open to Relocation

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on expertise in LLMs, RAG workflows, and agentic AI using GCP and AWS. Developed a semantic search engine using **42,549 arXiv papers** with a query rate of **2,979 QPS**, and implemented a clinical RAG workflow that provides evidence-based responses through FastAPI. Experienced researcher in genomics ML modeling (GWAS/SNP) with an accuracy of **87.6%** for predicting MS risk. Skilled in end-to-end ML operations—data ingestion, training, packaging, and CI/CD. i am Willing to relocate

Technical Skills

Generative AI & LLMs: RAG Systems, Prompt Engineering, LangChain, LlamaIndex, Hugging Face Transformers, LLaMA-3, BERT, RAG Evaluation (RAGAS)

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, RNNs, Ensemble Learning, Hyperparameter Tuning, Transfer Learning

MLOps & Deployment: Docker, FastAPI, CI/CD (GitHub Actions), MLflow, DVC, Model Deployment, REST APIs

Data Engineering & Processing: PySpark, Pandas, Polars, NumPy, SQL (Advanced), PostgreSQL, Data Pipelines

Cloud & Infrastructure: AWS (EC2, S3), GCP (Vertex AI, GCS), Hugging Face Spaces

Languages: Python (Advanced), Java, C++

Professional Experience

AI/ML Engineer Intern · NASO Technologies

Chennai, India | Dec 2024 – May 2025

- Developed **Well.AI** medical artificial intelligence application with an RAG architecture (LangChain, Hugging Face, ChromaDB, FastAPI) on GCP that facilitates medical query processing through machine learning
- Minimized hallucinations and maximized factualness in medical QA tasks by enhancing prompt and retrieval pipeline design to boost performance
- Designed and deployed microservices via Docker with structured logging for traceability and audit-ready ML applications via Supabase
- Worked with healthcare specialists to optimize data chunking and retrievals to achieve clinical performance through ML systems

Machine Learning Researcher · RUSA Project, University of Madras

Chennai, India | Jan 2025 – Aug 2025

- Created end-to-end machine learning pipeline for genomic risk prediction using ensemble techniques (XGBoost, Random Forest) resulting in an accuracy rate of **87.6%** on GWAS data for Multiple Sclerosis
- Created an interactive dashboard with Streamlit that allows non-technical users to conduct analysis of the results without requiring any coding knowledge
- Integrated experiment logging and data versioning using DVC on Google Cloud Platform (GCP)
- Utilized machine learning on high-dimensional biological data and bridged the gap between science and production ML engineering

Key Projects

PaperLens — Semantic Search & RAG System — Sentence Transformers, FAISS, FastAPI, LLaMA-3, Docker, Streamlit

- Built a high-performance semantic search engine indexing **42K+ research papers** across 20 domains, supporting real-time retrieval at **3K QPS with sub-ms latency**
- Engineered embedding cache reducing query latency from **1.1s to <1ms (55× improvement)**; exposed scalable REST APIs for search and summarization
- Designed end-to-end RAG pipeline generating structured research summaries using LLMs; deployed as an interactive web application

Enterprise Ticket Intelligence System — Agentic RAG — FastAPI, ChromaDB, LangChain, LLM Agents, Streamlit

- Developed an agent-driven RAG system to analyze **70MB+ enterprise support data**, enabling natural language querying for non-technical users
- Reduced dependency on manual analytics and SQL workflows by introducing automated insights and contextual responses
- Achieved **sub-200ms retrieval latency** with scalable document processing and multi-step reasoning agents

Brain Tumor Classification System — Medical AI Pipeline — TensorFlow, CNN, Transfer Learning, Docker

- Built an end-to-end medical imaging pipeline for MRI classification achieving **93% accuracy** across 4 tumor categories
- Implemented automated preprocessing and model inference pipeline with containerized deployment for reproducibility

EDUCATION

Master of Computer Applications (MCA) · Specialisation: ML, Deep Learning, Data Science
University of Madras, Chennai

GPA: 7.38/10 | 2023–2025

B.Sc. Biotechnology · Foundation in genomics, statistics, and computational biology
RKM Vivekananda College, Chennai

GPA: 7.2/10 | 2020–2023

ACHIEVEMENTS

- 2st Place**—AI Front-End Hackathon 2025, MCC Chennai (competed against teams from 25 colleges)
- Published technical articles** on EDA techniques, LLM finetuning, case study using ML, and Genomics ML on Medium
- Vice President**, Rotaract Club—led 5+ AI awareness & technology adoption events across Chennai
- Mentored **100+ students** in PCBM subjects, incorporating AI-driven learning approaches to improve conceptual understanding and engagement